Atul Kumar

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Education

Indian Institute of Information Technology Design and Manufacturing (IIITDM) Kurnool Kurnool, India
Bachelor of Technology in Artificial Intelligence and Data Science; CGPA: 8.16/10 *Aug 2024 – May 2028 (Expected)*

Relevant Coursework: Data Structures and Algorithms, Machine Learning, Linear Algebra, Probability and Statistics, Object-Oriented Programming, DBMS

Technical Skills

Programming Languages: Python, C++, C, SQL

Machine Learning: Supervised and Unsupervised Learning, Linear and Logistic Regression, Decision Trees, Random Forest, XGBoost, SVM, KNN, Naive Bayes, K-Means, DBSCAN, PCA, Cross-Validation, Hyperparameter Tuning, Model Evaluation

Deep Learning: ANN, CNN, RNN, LSTM, Attention Mechanisms, Transformers, Transfer Learning, Backpropagation

Computer Vision & LLM Systems: Semantic Segmentation, Cross-Modal Fusion, Image Captioning, Retrieval-Augmented Generation (RAG), Corrective RAG

Frameworks & Libraries: PyTorch, TensorFlow, Keras, LangGraph, ChromaDB, Ragas, FastAPI, Scikit-learn, NumPy, Pandas, Matplotlib, Streamlit

Data & Tools: Exploratory Data Analysis, Data Cleaning, Data Preprocessing, Feature Engineering, Git, GitHub, Linux, DVC, Streamlit Cloud

Projects

RGB–Thermal Semantic Segmentation | *PyTorch, ResNet-18, U-Net, Cross-Modal Attention, AMP* Sep 2026 – Oct 2026

- Engineered a **29.06M-parameter** dual-stream ResNet-18 network for multi-spectral segmentation on the **MFNet dataset (1,568 four-channel image pairs)**.
- Designed a 5-scale Cross-Modal Attention Fusion (CMAF) module pairing channel SE-gating with spatial attention, routing multimodal features into a U-Net decoder.
- Achieved **55.78% validation mIoU**, **50.60% foreground mIoU**, and **97.11% pixel accuracy**, improving by 21.22 percentage points over the 34.56% scratch baseline. **GitHub:** Repository

Self-RAG on Indian Constitution & IPC | *LangGraph, DeepSeek R1/V3, ChromaDB, Ragas, FastAPI* Aug 2026 – Oct 2026

- Architected an agentic legal question-answering system over Constitution articles and IPC sections using a 16-node **LangGraph** workflow, DeepSeek query routing, and parallel **ChromaDB** vector retrieval.
- Implemented corrective retrieval featuring parallel relevance scoring, an automated hallucination guard with critic revision, an answer usefulness judge, and web fallback.
- Evaluated on a 50-example synthetic test set, achieving **0.96 Faithfulness**, **0.95 Context Precision**, and **0.90 Context Recall** with Ragas; served via **FastAPI**. **GitHub:** Repository

Image Captioning System using Deep Learning | *TensorFlow, Keras, InceptionV3, LSTM, Attention* Jan 2026 – Mar 2026

- Developed an end-to-end captioning pipeline on **Flickr8k (8,091 images, 40,455 captions)**, engineering a vocabulary preprocessing pipeline with a custom tokenizer (2,827 tokens, 34-token cap).
- Integrated **Bahdanau additive attention** over unpooled $8 \times 8 \times 2{,}048$ InceptionV3 spatial feature maps (64 grid vectors) to preserve image-region information.
- Evaluated the CNN–LSTM baseline at **BLEU-1 0.4830** and **BLEU-4 0.1012**; deployed a cached **Streamlit** web application for real-time inference. **GitHub:** Repository

Activities

Smart India Hackathon (SIH) Kurnool, India
Team Lead *2025*

- Led a 6-member team to develop a functional prototype during a 36-hour hackathon.
- Presented the working prototype to judges.

RizeOS Hackathon
Team Lead

- Led a 6-member team during the RizeOS Hackathon.

Achievements & Coding Profiles

Achievements: Secured **2nd place** in RizeOS Hackathon | Achieved **team rank 1,500** in ICPC Preliminary Round.

Solved **585+** data structures and algorithms problems across LeetCode, GeeksforGeeks, and CodeChef. Attained **CodeChef 2-Star** (1597, 24 rated contests) and **LeetCode rating 1653**; top 50 at IIITDM Kurnool on GeeksforGeeks.

Coding Profiles: LeetCode | CodeChef | Codeforces